BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 029 · Standards and Protocols

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What is SMTP used for?

Q2. Unsigned 8-bit values can reach 255. Does 248 + 16 overflow?
A. Yes: the sum is 264
B. No: the sum is 264
C. No: the sum is 8
D. Yes: the sum is 232

Q3. What is the Internet?

Q4. A browser asks for a school's web page. “A client must host a website before asking for it.” Correct this. Which computer provides the page?

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding

I Do · Read new knowledge

Key vocabulary

Standard — Agreed technical specification allowing compatibility.

Protocol — Agreed rules for communication between devices.

Interoperability — Different hardware/software systems working together.

HTTP — Protocol for web communication.

TCP — Supports ordered/reliable data transport.

IP — Supports addressing and routing.

Layering — Separating communication tasks into different functions.

New learning

Why standards matter: Standards are agreed rules/specifications that allow equipment and software from different manufacturers to communicate.

Protocols in action: A protocol provides rules for exchanging data. Different services need different protocol functions.

Different communication jobs: A web request uses a web protocol such as HTTP/HTTPS. TCP supports reliable ordered delivery, while IP handles addressing and routing.

Layering is an explanation: Communication tasks can be considered in stages/layers. OCR J277 does not require memorising names and functions of all TCP/IP model layers; focus on the required standards and protocol functions.

Worked example

A school purchases equipment from several manufacturers. Explain why standards and communication protocols help it work together. [4 marks]

1. Different manufacturers need compatible rules if their devices are to communicate.

2. Agreed standards [1] help equipment from different makers interoperate [1].

3. Communication protocols define common rules for sending data [1], so requests can be interpreted correctly by devices [1].

4. Four points: standard → compatibility, protocol → effective exchange.

We Do · Practise with guidance

Guided 1

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

correct order [1]. IP supplies destination addressing/routing [1].

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Sending email from a client to a mail server and between mail servers.

Q2. A — Yes: the sum is 264

Q3. A worldwide collection of interconnected computer networks.

Q4. A web server stores website resources and sends them when clients request them. Access needs working servers and network services.

I Do / We Do pages · Guided practice

Q1. Both devices follow agreed networking rules [1] even though built by different manufacturers [1], so they can exchange compatible data [1] and use network services [1].

Q2. HTTP sets rules for requesting/transferring web content [1] so client/server know which resource is needed [1]. TCP helps deliver data reliably [1] in correct order [1].

Q3. TCP supports reliable delivery [1] and correct ordering [1]; IP uses destination network addressing [1] and routing between networks [1].

You Do pages · Code First

```
print("Request")
print("Response")
```

Expected result: Request Response

You Do pages · Independent questions

Q1. They allow equipment/software from different manufacturers to communicate/interoperate [1].

Q2. Both systems use the same agreed message rules [1], so requests/responses are understood despite different manufacturers [1]. Shared standards enable interoperability [1], avoiding a separate custom communication system for each manufacturer's equipment [1].

Q3. Standards support device compatibility [1] between different manufacturers [1]. HTTP defines web requests and responses [1]. TCP supports reliable arrival [1] and

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

A tablet from one company uses a website hosted by another. Explain why communication standards matter. [4 marks]

Worked method: Both devices follow agreed networking rules [1] even though built by different manufacturers [1], so they can exchange compatible data [1] and use network services [1].

Guided 2

A client requests a web page. Explain the separate roles of HTTP and TCP. [4 marks]

Worked method: HTTP sets rules for requesting/transferring web content [1] so client/server know which resource is needed [1]. TCP helps deliver data reliably [1] in correct order [1].

Guided 3

Explain how TCP and IP work together when sending data between networks. [4 marks]

Worked method: TCP supports reliable delivery [1] and correct ordering [1]; IP uses destination network addressing [1] and routing between networks [1].

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Write the second print instruction.

```
print("Request")
```

write

Question 1

State one benefit of agreed networking standards. [1 mark] Source: Adapted from your Paper 1 revision booklet, 1.3.2.

Question 2

A public-transport ticket kiosk communicates with a booking service made by a different manufacturer. Explain two benefits of both systems following agreed communication standards or protocols. [4 marks] Source: Original J277-style question; not an OCR past-paper original.

Question 3

Explain how standards and HTTP, TCP and IP help successful network communication. [6 marks] Source: Original J277-style question; not an OCR past-paper original.

BLACK PEN Answer all questions first. GREEN PEN Then: tick correct answers; rewrite mistakes.

Exit Ticket · Answer before checking

OCR J277 requires memorising all four TCP/IP layer names and their functions.